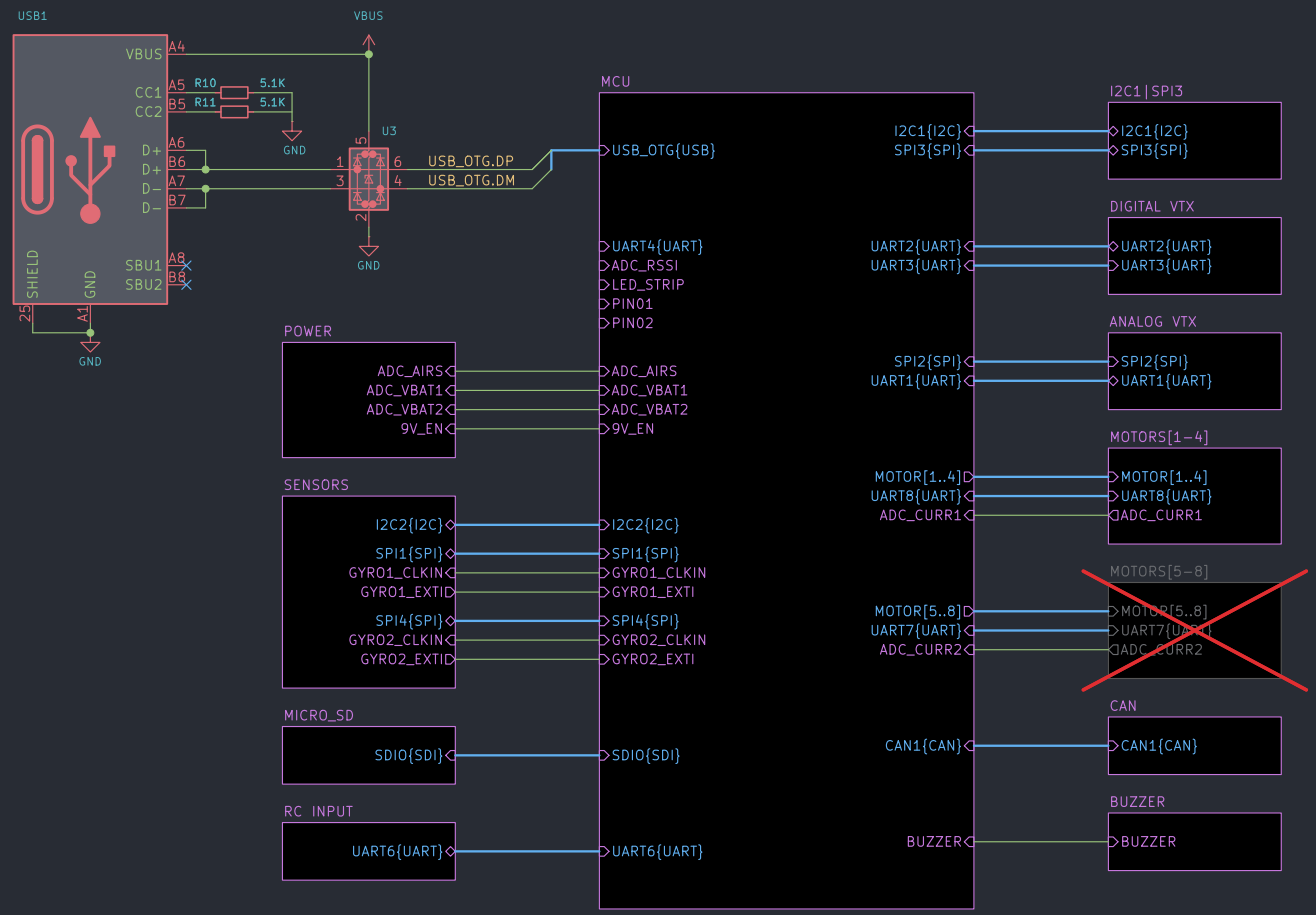
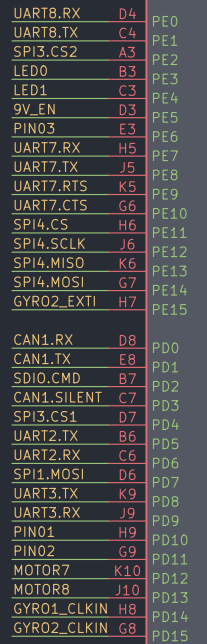
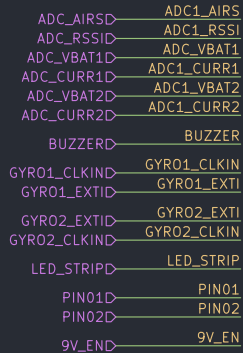
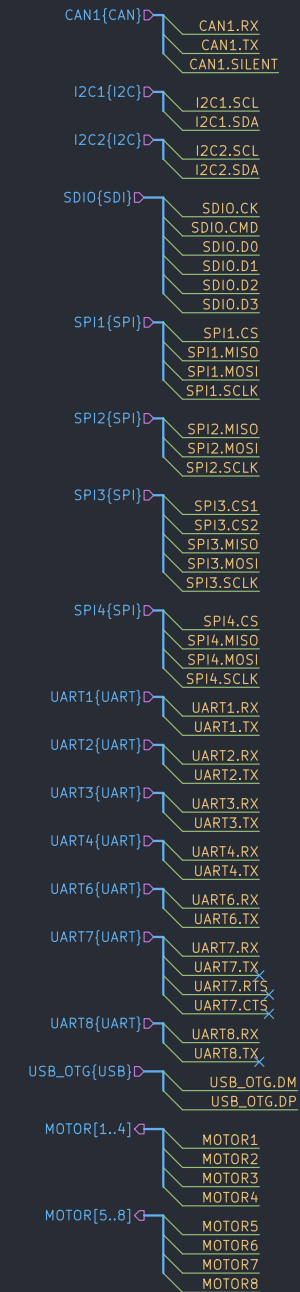


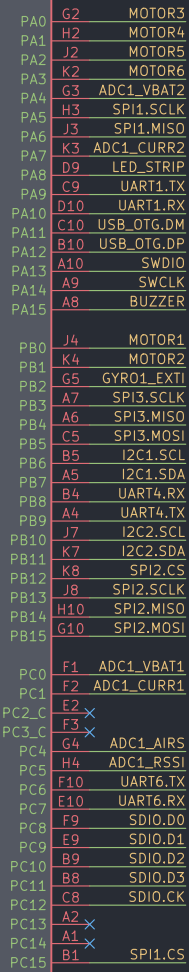
LEVIA H7



MCU



U2A
STM32H743VIHx



ADC
PC0 BATT_VOLTAGE_SENS ADC1 SCALE(1)
PC1 BATT_CURRENT_SENS ADC1 SCALE(1)
PA4 BATT2_VOLTAGE_SENS ADC1 SCALE(4)
PA7 BATT2_CURRENT_SENS ADC1 SCALE(1)



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Sheet: /MCU/

File: MCU.kicad_sch

Title: LEVIA H7

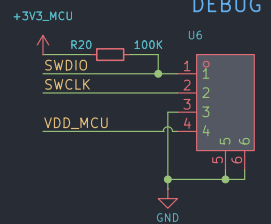
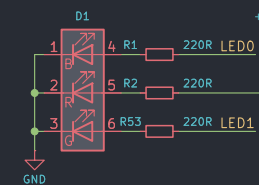
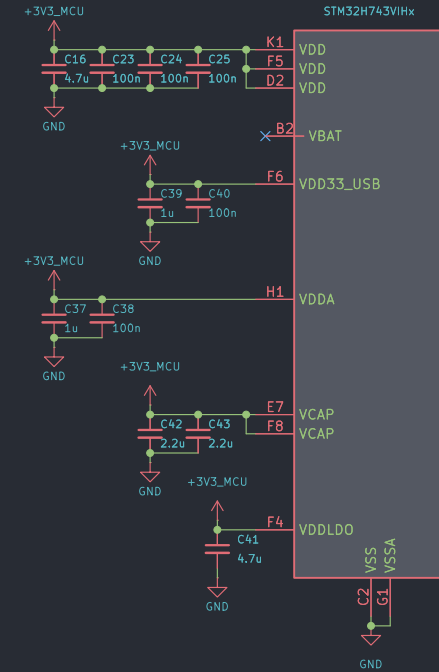
Size: A4

Date: 2026-06-13

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Rev: 0.1-prelayout

Id: 2/13

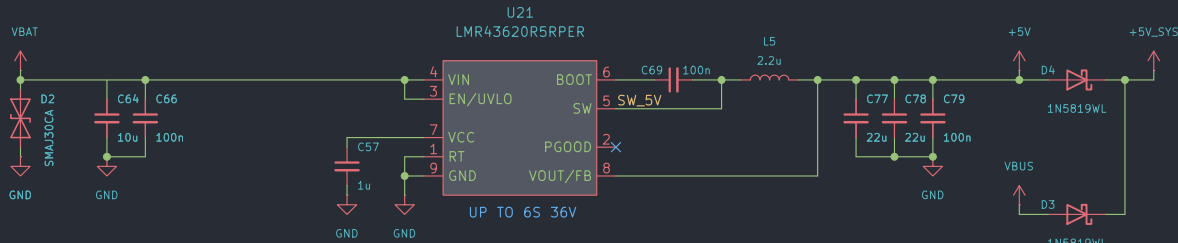


DEBUG

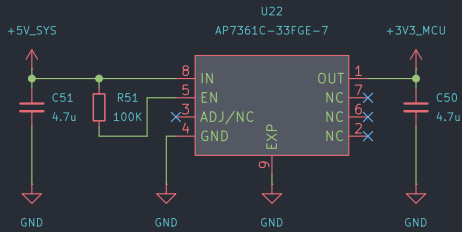
POWER

9V_EN → 9V_EN
ADC_VBAT1 → ADC_VBAT1
ADC_VBAT2 → ADC_VBAT2
ADC_AIRSD → ADC_AIRSPEED

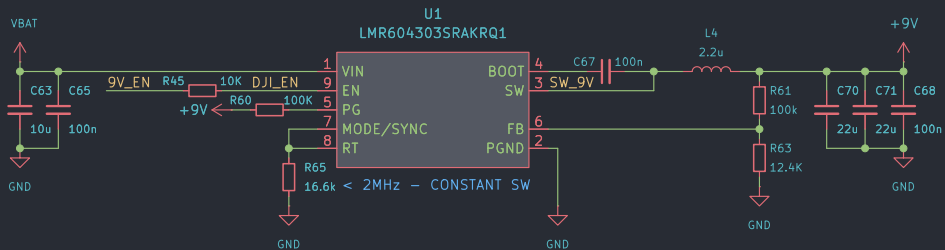
+5V - MCU + SENSORS VIA LDOs



+3V3 - MCU

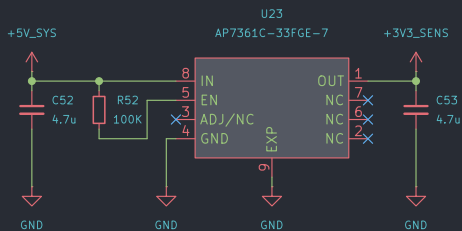


+9V - DJI VIDEO TX

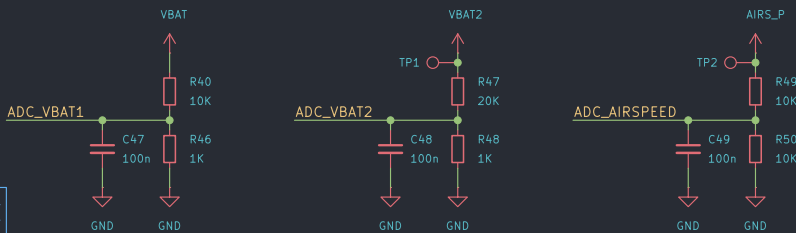


PIN-TO-PIN COMPATIBLE WITH 4A/6A VERSIONS LIKE LMR60440-Q1

+3V3 - SENSORS



ANALOG SIGNALS



VBAT1: 10k / 1k → 11:1
VBAT2: 20k / 1k → 21:1
AIRS: 10k / 10k → 2:1

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Sheet: /POWER/

File: POWER.kicad_sch

Title: LEVIA H7

Size: A4

Date: 2026-06-13

KiCad E.D.A. 10.0.3

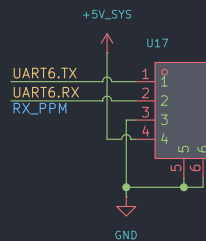


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Id: 3/13

RC INPUT

UART6{UART} ◊
UART6.RX
UART6.TX



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Sheet: /RC INPUT/

File: RC_INPUT.kicad_sch

Title: LEVIA H7

Size: A5

Date: 2026-06-13

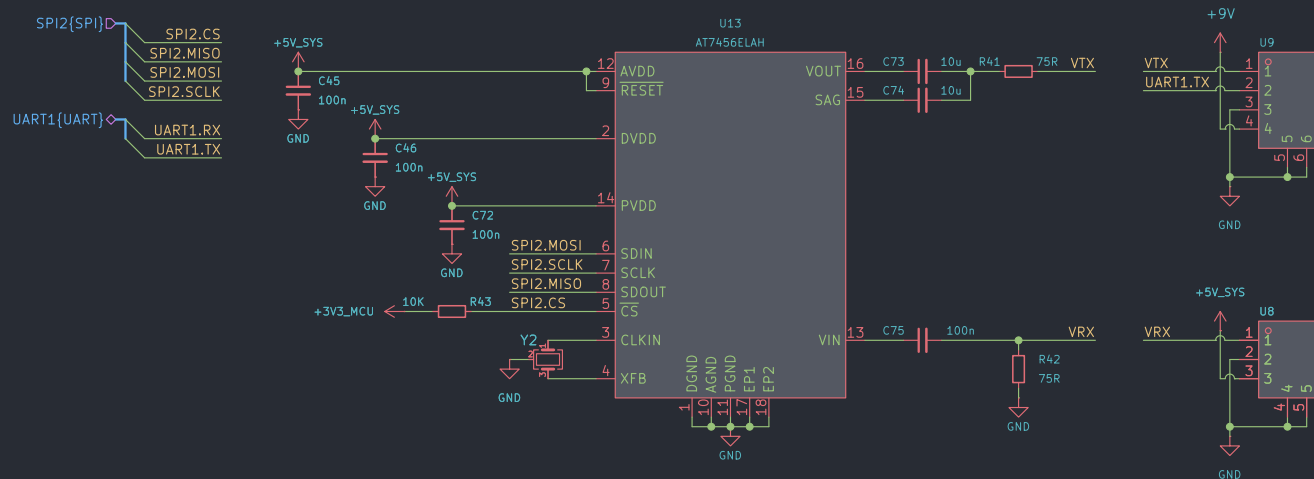
KiCad E.D.A. 10.0.3

Rev: 0.1-prelayout

Id: 4/13



ANALOG VTX



FW NOTES:

https://raw.githubusercontent.com/ArduPilot/ardupilot/master/libraries/AP_HAL_ChibiOS/hwdef/MatekH743/hwdef.dat

```
# SPI2 for MAX7456 OSD
PB12 MAX7456_CS CS
PB13 SPI2_SCK SPI2
PB14 SPI2_MISO SPI2
PB15 SPI2_MOSI SPI2
```

OSD ALTERNATIVES

! TO BE TESTED !

MAX7456EUI:	C5346361
AT7456ELAH:	C42388749

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Sheet: /ANALOG VTX/
File: ANALOG_VTX.kicad_sch

Title: LEVIA H7

Size: A5 Date: 2026-06-13
KiCad E.D.A. 10.0.3

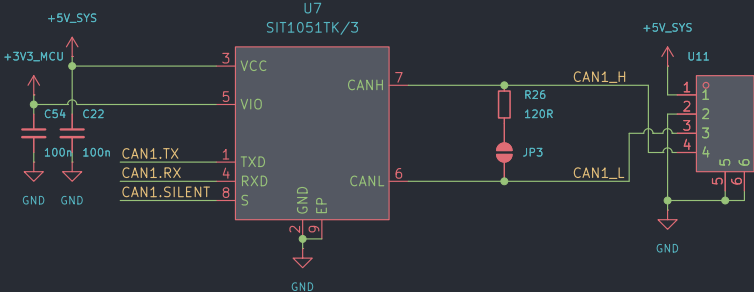
Rev: 0.1-prelayout
Id: 5/13



CAN

CAN1{CAN}D

- CAN1.RX
- CAN1.TX
- CAN1.SILENT



CAN-FD ALTERNATIVES
! TO BE TESTED !
TJA1051TK/3,118: C124020
SIT1051TK/3: C1121846

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Sheet: /CAN/
File: CAN.kicad_sch

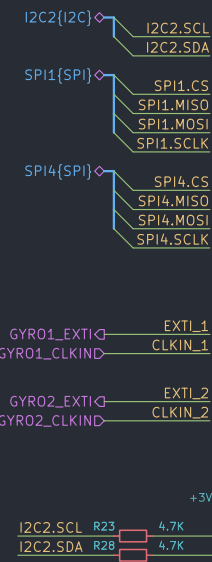


Title: LEVIA H7

Size: A5 | Date: 2026-06-13
KiCad E.D.A. 10.0.3

Rev: 0.1-prelayout
Id: 6/13

SENSORS

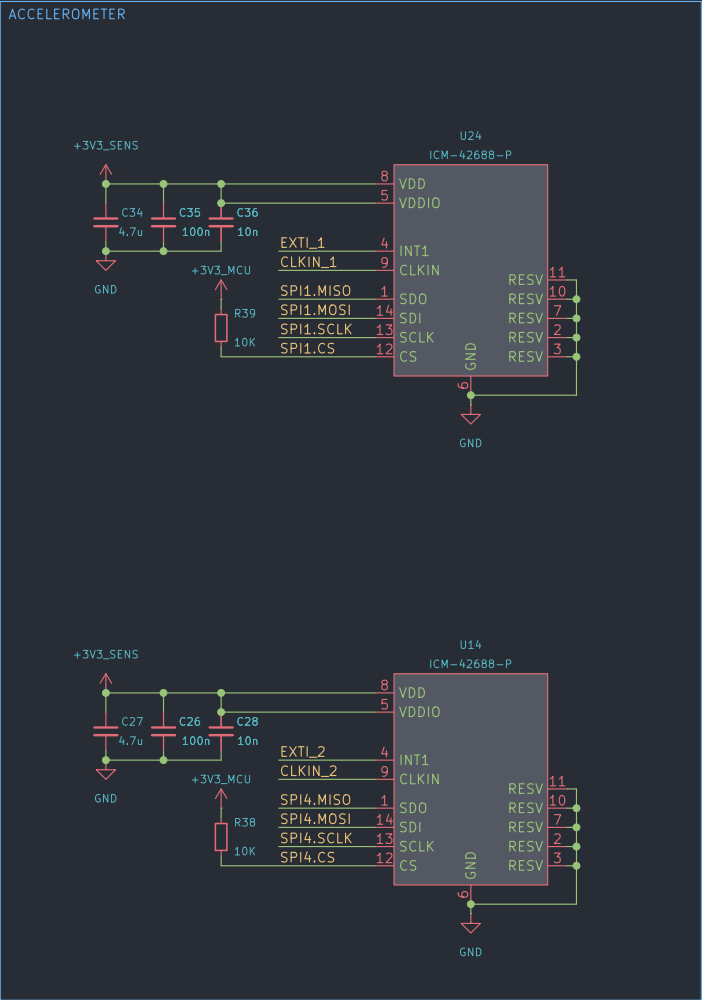


FW NOTES:

https://raw.githubusercontent.com/ArduPilot/ardupilot/master/libraries/AP_HAL_ChibiOS/hwdef/MatekH743/hwdef.dat

SPI1 for IMU1
PA5 SPI1_SCK SPI1
PA6 SPI1_MISO SPI1
PD7 SPI1_MOSI SPI1
PC15 IMU1_CS CS

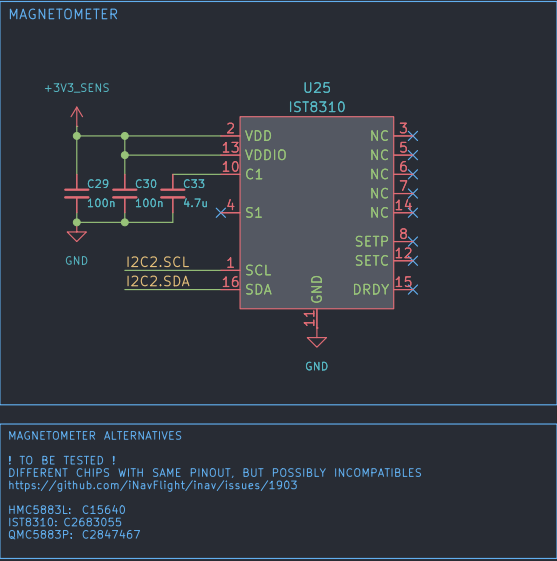
SPI4 for IMU2
PE11 IMU2_CS CS
PE12 SPI4_SCK SPI4
PE13 SPI4_MISO SPI4
PE14 SPI4_MOSI SPI4



ACCELEROMETER ALTERNATIVES

! TO BE TESTED !

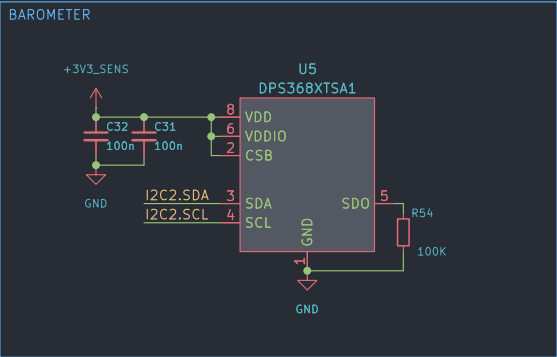
ICM-42688-P: C1850418
ICM-42688-P(TOKMAS): C54308212



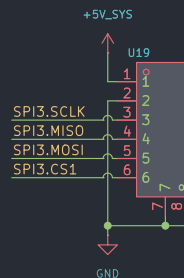
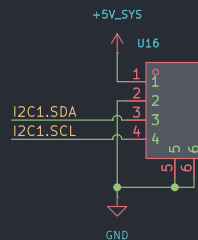
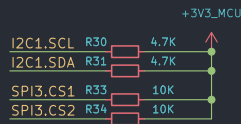
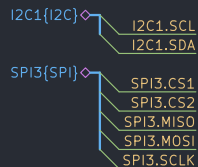
MAGNETOMETER ALTERNATIVES

! TO BE TESTED !
DIFFERENT CHIPS WITH SAME PINOUT, BUT POSSIBLY INCOMPATIBLES
<https://github.com/ArduPilot/ardupilot/issues/1903>

HMC5883L: C15640
IST8310: C2683055
QMC5883P: C2847467



I2C1|SPI3



! MIND THE VOLTAGE !

POWER PIN: +5V
LOGIC: +3.3 V

FW NOTES:

https://raw.githubusercontent.com/ArduPilot/ardupilot/master/libraries/AP_HAL_ChibiOS/hwdef/MatekH743/hwdef.dat

SPI3 - external
PB3 SPI3_SCK SPI3
PB4 SPI3_MISO SPI3
PB5 SPI3_MOSI SPI3

external CS pins
PD4 EXT_CS1 CS
PE2 EXT_CS2 CS

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Sheet: /I2C1|SPI3/

File: I2C1|SPI3.kicad_sch

Title: LEVIA H7

Size: A5

Date: 2026-06-13

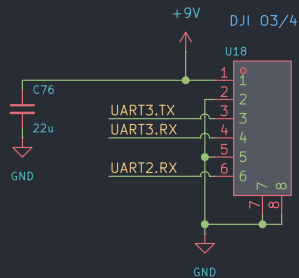
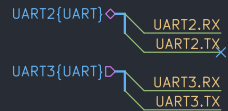
KiCad E.D.A. 10.0.3

Rev: 0.1-prelayout

Id: 8/13



DIGITAL VTX



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Sheet: /DIGITAL VTX/

File: DIGITAL_VTX.kicad_sch

Title: LEVIA H7

Size: A5

Date: 2026-06-13

KiCad E.D.A. 10.0.3

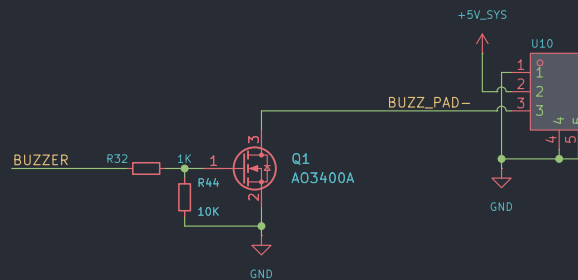
Rev: 0.1-prelayout

Id: 9/13



BUZZER

BUZZERD BUZZER



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Sheet: /BUZZER/

File: BUZZER.kicad_sch

Title: LEVIA H7

Size: A5

Date: 2026-06-13

KiCad E.D.A. 10.0.3

Rev: 0.1-prelayout

Id: 10/13

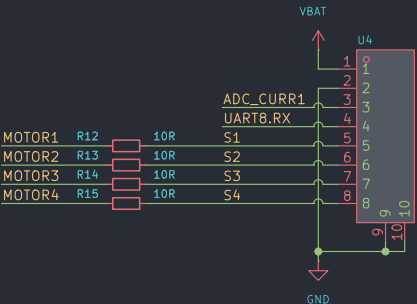
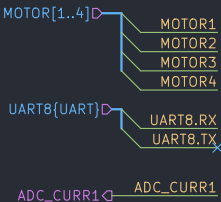


[illegible]

Id: 11/13



MOTORS [1-4]



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Sheet: /MOTORS[1-4]/
File: MOTORS[1-4].kicad_sch

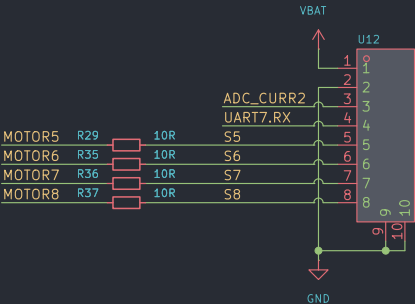
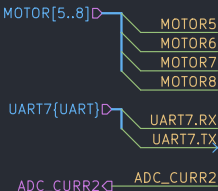


Title: LEVIA H7

Size: A5 Date: 2026-06-13
KiCad E.D.A. 10.0.3

Rev: 0.1-prelayout
Id: 12/13

MOTORS [5-8]



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Sheet: /MOTORS[5-8]/
File: MOTORS[5-8].kicad_sch



Title: LEVIA H7

Size: A5
KiCad E.D.A. 10.0.3

Date: 2026-06-13

Rev: 0.1-prelayout
Id: 13/13